

Walk-unbiased doubly robust graph propagation for recommendation from exposure-biased logs

Abstract

Graph recommenders propagate preferences along multi-hop walks of a logged graph whose edges exist only where an item was exposed. Propagation-level debiasing re-weights these edges by inverse propensities or completes them with doubly robust (DR) estimates. We show that such corrections are biased for multi-hop propagation: a walk that traverses an edge several times multiplies that edge’s estimate with itself. We give the exact bias, whose worst case grows as the inverse of the propensity clip, and remove it with DRUP (Doubly Robust, walk-Unbiased Propagation), which replaces every repeated edge by its idempotent unbiased counterpart. Möbius inversion over walk-index coincidences makes the correction exact for any odd number of hops at the cost of the uncorrected operator. With fixed nuisances, DRUP is edge-wise doubly robust for propagation over the full-exposure graph, its bias under within-user exposure dependence is of second order in the imputation error, and its expected scores do not respond to interventions on exposure. As a linear operator it also admits exact attributions, certificates against injected profiles and a jointly private release, within a stated scope. On Coat, Yahoo! R3 and KuaiRec, against 22 tuned methods, the correction leaves top- K accuracy unchanged. Training-free propagation over the DR graph is the most accurate method on KuaiRec (fully observed test matrix)

while graph models trained with pairwise losses are better on Yahoo!R3. Real-data interventions confirm exposure invariance on Coat and KuaiRec; on the sparse Yahoo!R3 log it requires small clips that cost accuracy.

Keywords: Causal recommendation, graph-based recommendation, doubly robust estimation, exposure bias, missing not at random, exposure invariance, trustworthy recommendation.

1. Introduction

Graph-based recommender systems represent users and items as nodes and logged interactions as edges, and they score a user–item pair by aggregating signals along multi-hop walks (He et al., 2020; Shen et al., 2021). The logged graph, however, is not the preference graph. An edge exists only if the platform *exposed* the item and the user then responded, so the graph records the exposure policy as much as the user’s taste. A recent survey of causal learning in graph-based recommendation (Yu et al., 2026) traces several open problems of the field, among them fairness, interpretability and out-of-distribution behaviour, to message passing over this observational graph, and it lists theoretical guarantees and privacy as open directions.

Two families of corrections exist. *Loss-level* debiasing re-weights the training objective by inverse propensities or doubly robust (DR) terms (Schnabel et al., 2016; Wang et al., 2019b; Saito et al., 2020), but it leaves the propagation operator of a graph model untouched. *Propagation-level* debiasing changes the propagated graph itself, for example by weighting each edge by its inverse propensity (Kim et al., 2022) or by learning popularity-aware aggregation weights (Zhou et al., 2023; Islam et al., 2026). This paper starts

from a simple observation about the second family. An inverse-propensity edge estimate $W_e = O_e Y_e / p_e$ is unbiased for the potential outcome Y_e , but a walk that traverses e twice contributes W_e^2 , and $\mathbb{E}[W_e^2] = Y_e / p_e \neq Y_e$. The same happens for DR edge estimates. Walks that repeat an edge are part of every multi-hop operator: the back-and-forth step $u \rightarrow j \rightarrow u$ of a three-hop walk is one. Their excess is largest where the propensity is smallest, that is, on the rarely exposed edges that debiasing is meant to recover.

We make this observation precise and remove the resulting bias exactly. Two points qualify its practical reach, and we state them up front. First, for ranking the *unexposed* candidates of a user with inverse-propensity edges, the repeated walks through the candidate vanish at three hops (the candidate’s own estimate is zero), and the bias only appears from five hops on; for DR edges it is present at three hops. Second, on the three public benchmarks we use, removing the bias does not change top- K accuracy. The value of the correction is therefore that the propagated scores become unbiased estimates of a well-defined counterfactual quantity, with guarantees that can be checked, rather than an accuracy gain.

Contributions.

1. *Diagnosis.* We give the exact conditional bias of inverse-propensity and DR graph propagation on unexposed candidates. It is unbounded as propensities vanish, and under a propensity clip τ its worst case is $\Theta(1/\tau)$ (Theorem 5).
2. *Estimator.* DRUP (Doubly Robust, walk-Unbiased Propagation) uses $Y_e^2 = Y_e$ for binary outcomes and replaces every repeated edge by its idempotent unbiased counterpart. The three-hop operator has the cost

of one item Gram matrix and factorises into a public item operator and per-user corrections. Möbius inversion over the coincidence patterns of walk indices makes the correction exact for any odd number of hops (Theorem 3).

3. *Guarantees.* With fixed nuisances DRUP is edge-wise doubly robust for propagation over the full-exposure graph (Theorem 1); on unexposed candidates it is unbiased up to the imputation error at the candidate itself, which no estimator can remove (Corollary 2); its bias under exposure dependence within a user’s history is of second order in the imputation error (Proposition 4); we bound its variance and clipping bias (Theorem 6); and its expected scores do not respond to interventions on an item’s exposure (Theorem 7).
4. *Structural properties.* As a linear operator with fixed nuisances, DRUP also admits exact attributions and optimal counterfactual explanations, closed-form certificates against injected profiles, a jointly private release of its item operator and an exposure-capped allocation. These properties follow from linearity and hold for any linear propagation operator, including the uncorrected ones; we state them with their scope (frozen or public nuisances) and report where they are vacuous.
5. *Evaluation.* On Coat (Schnabel et al., 2016), Yahoo!R3 (Marlin and Zemel, 2009) and KuaiRec (Gao et al., 2022), whose test data are random or fully observed, we compare 23 methods under one protocol: cross-fitted nuisances, per-split validation for every method, and user-level significance tests with Holm correction. We check each guarantee by Monte Carlo, including stress tests outside its assumptions, and by

real-data interventions on exposure.

2. Related work

We review six lines of work that DRUP touches. Table 1 summarises how DRUP relates to representative methods.

Graph-based recommendation. Graph neural recommenders learn user and item representations by propagating embeddings over the user–item graph. NGCF (Wang et al., 2019a) stacks non-linear propagation layers, and LightGCN (He et al., 2020) shows that removing feature transformations and non-linearities improves accuracy, which reduces propagation to a weighted sum of powers of a normalised adjacency. UltraGCN (Mao et al., 2021) approximates infinite-layer propagation with a constraint loss, and SGL (Wu et al., 2021) and SimGCL (Yu et al., 2022) add contrastive views. Closed-form models are strong competitors of trained propagation: EASE (Steck, 2019) learns an item–item matrix with a zero diagonal in closed form, GF-CF (Shen et al., 2021) and BSPM (Choi et al., 2023) interpret collaborative filtering as graph signal filtering, and graph convolutions can trade accuracy for diversity (Isufi et al., 2021). Reproducibility studies show that carefully tuned simple baselines often match graph models (Anelli et al., 2023). All these models propagate over the *logged* graph. DRUP keeps the linear propagation form of LightGCN and of closed-form filters but propagates over an unbiased estimate of the full-exposure graph, and we compare it with EASE and GF-CF on both graphs.

Bias and debiasing in recommendation. Recommendation logs are missing not at random: users see what the system exposed and rate what they chose

to rate (Marlin and Zemel, 2009). Surveys organise the resulting biases and their remedies (Chen et al., 2023) and cast debiasing as causal inference (Gao et al., 2024). Propensity-based methods re-weight the loss by inverse exposure probabilities (Schnabel et al., 2016), possibly self-normalised to reduce variance (Swaminathan and Joachims, 2015); Saito et al. (2020) extend them to implicit feedback, and Liang et al. (2016) model exposure as a latent variable. Doubly robust learning combines inverse propensities with an imputation model (Wang et al., 2019b), and later variants reduce its bias and variance (Guo et al., 2021; Dai et al., 2022; Li et al., 2023) or learn information-theoretic representations (Wang et al., 2020). A small uniformly logged sample can supervise the debiasing itself (Chen et al., 2021). Causal approaches to popularity bias remove its direct effect by counterfactual inference (Wei et al., 2021), adjust for it as a confounder (Zhang et al., 2021), or disentangle interest from conformity (Zheng et al., 2021). In this journal, popularity debiasing has been studied from the user and item side jointly (Boratto et al., 2021), with a controllable accuracy trade-off (Okamura et al., 2024), through invariant learning with biased imputation (Bai et al., 2025), and by causal deconfounding of multi-criteria ratings (Guo et al., 2026). These methods correct the loss or the output of a model. DRUP applies IPS and DR estimation to the *edges* of the propagated graph, and its control-variate family interpolates between the two.

Debiasing inside graph propagation. Closest to our work are methods that change neighbour aggregation. NAVIP (Kim et al., 2022) weights each edge by its inverse propensity inside LightGCN. r-AdjNorm (Zhao et al., 2022) shows that the normalisation exponent of the adjacency controls how strongly

popularity is amplified, and tunes it. APDA (Zhou et al., 2023) learns an adaptive inverse-popularity weight per edge, CAGED (Que et al., 2025) derives aggregation weights from a backdoor adjustment, and DPAA (Islam et al., 2026) learns interaction- and layer-wise weights to counter the amplification of popularity by repeated message passing. These methods change *which* adjacency is propagated and target popularity-related metrics. None of them analyses what happens to an unbiased edge estimate when the re-weighted adjacency is multiplied with itself over several hops. Theorem 5 shows that the product is then biased, by a term of order $1/p$ that concentrates on rarely exposed items, and DRUP removes this term exactly for any number of hops. Our linear propagation baseline over the logged graph searches the normalisation exponent and is therefore a training-free r-AdjNorm; we also train r-AdjNorm and NAVIP.

Unbiased estimation of polynomials under missingness. The walk correction belongs to a classical idea: an unbiased estimator of a polynomial in unknown quantities must avoid products of the same noisy measurement, as in U-statistics (Hoeffding, 1948). For covariance matrices estimated from Bernoulli-masked data, the off-diagonal entries of the masked Gram matrix need a $1/p^2$ correction and the diagonal a $1/p$ correction (Lounici, 2014); our diagonal-corrected item operator G is the analogue for inverse-propensity and DR edges with heterogeneous propensities. Unbiased subgraph counts from sampled edges (Tsourakakis et al., 2009), and the passage from homomorphism counts to injective counts by Möbius inversion on the partition lattice (Rota, 1964), are the combinatorial counterparts of our K -hop construction. What is new here is the combination of these tools with DR edge

estimates and degree weights for recommendation, the public-operator form, and the consequences for exposure invariance. Our treatment of nuisances follows double machine learning, whose cross-fitting we adopt (Chernozhukov et al., 2018), and our clipping and control-variate choices mirror doubly robust off-policy evaluation (Dudík et al., 2011).

Fairness in recommendation. Fairness notions in recommendation cover users and items, groups and individuals, and outcomes and exposure (Wang et al., 2023); a unifying account of their measurement is given by Amigó et al. (2023), and provider fairness and its relation to data characteristics have been analysed in Gómez et al. (2022); Deldjoo et al. (2021). On the item side, exposure can be allocated in proportion to merit (Singh and Joachims, 2018), amortised over time (Biega et al., 2018) or under uncertainty about merit (Singh et al., 2021), and two-sided allocations can be guaranteed by constrained matching (Patro et al., 2020). Fairness can also be audited by counterfactual reasoning (Cornacchia et al., 2023). DRUP contributes a causal item-side property, *exposure invariance*: its expected scores do not respond to interventions on an item’s exposure (Theorem 7). Exposure invariance is a necessary condition for merit-based exposure fairness, not a distributive criterion: it removes the dependence of scores on exposure history but can increase the concentration of recommendations. The exposure-capped allocation (Theorem 12) is the distributive mechanism; like the matching view of Patro et al. (2020) it is a b -matching.

Explanations, robustness and privacy. Explainable recommendation is surveyed by Zhang and Chen (2020). For graph models, GNNExplainer (Ying et al., 2019) and CF-GNNExplainer (Lucic et al., 2022) search for important

or minimal sets of edges, PRINCE (Ghazimatin et al., 2020) finds minimal sets of a user’s actions whose removal changes a random-walk recommendation, and CountER (Tan et al., 2021) optimises counterfactual explanations of aspect-based models. Counterfactual explanations have a legal motivation (Wachter et al., 2018) and hidden assumptions (Barocas et al., 2020), and Shapley attributions depend on their baseline (Kumar et al., 2020). Recommenders can be manipulated by injected profiles (Lam and Riedl, 2004; Li et al., 2016; Fang et al., 2018), and shilling groups can be detected with graph convolutions (Wang et al., 2022); PORE (Jia et al., 2023) certifies robustness by ensembling models trained on sub-sampled data. Private recommenders perturb item–item statistics (McSherry and Mironov, 2009), factorisation models (Liu et al., 2024) or matrix completion under joint differential privacy (Jain et al., 2018), and federated approaches keep ratings on the device (Chai et al., 2021). Because a DRUP score with fixed nuisances is linear in each user’s own interactions, its attributions are exact and its minimal counterfactual explanations are found by sorting (Proposition 8, Theorem 9); the same linearity gives closed-form certificates against injected profiles (Theorem 10) and a jointly private release of one item operator with an analytically calibrated Gaussian mechanism (Balle and Wang, 2018; Hsu et al., 2014) (Theorem 11). These properties hold for any linear propagation operator.

3. Problem setting

Users $u \in [m]$, items $i \in [n]$ and pairs $e = (u, i)$ carry a binary potential outcome $Y_e \in \{0, 1\}$ (whether u would like i if it were shown) and an exposure

Table 1: DRUP and representative related methods. “loss”: the correction acts on the training objective; “output”: it acts on the predictions; “learned”: learned aggregation weights; “lin.”: follows from linearity with fixed nuisances and holds for any linear propagation operator (including linear LightGCN, GF-CF and DR adjacency); “—”: not applicable.

Method	debases propagation	unbiased beyond 1 hop	doubly robust	exposure invariance	exact attributions	certificates / joint DP
LightGCN, NGCF, UltraGCN, SGL, SimGCL	no	no	no	no	no	no
linear LightGCN, EASE, GF-CF, BSPM	no	no	no	no	lin.	lin.
IPS, SNIPS, DR, MRDR, StableDR	loss	—	DR variants	no	no	no
MACR, PDA, DICE	output	—	no	no	no	no
NAVIP	edge weights	no	no	no	no	no
r-AdjNorm, APDA, CAGED, DPAA	learned	no	no	no	no	no
GNNE explainer, CF-GNNE explainer	—	—	—	—	approx.	no
PORE	—	—	—	—	—	certified
DP / federated MF	—	—	—	—	—	privacy
DRUP (ours)	DR edges	any odd K	yes	in expectation*	lin.	lin.[†]

*Under Assumptions 1–2 and unclipped propensities (Theorem 7). [†]Frozen nuisances for certificates; public nuisances for joint DP.

indicator $O_e \in \{0, 1\}$. The log reveals O and $O \odot Y$.

Assumption 1 (Conditionally independent exposure). *Conditionally on Y , the exposures O_e are independent Bernoulli variables with propensities $p_e = \Pr(O_e = 1 \mid Y) > 0$.*

The propensity may depend on the pair’s own outcome, so the log may be missing not at random. All expectations below are over O given Y .

Assumption 2 (Fixed nuisances). *The propensity estimates $\hat{p}$, the imputation $\hat{Y} \in [0, 1]^{m \times n}$ and the edge weights C are fixed given Y , i.e., they do not depend on O (for instance, they are fitted on an independent log). We use*

$C_{ui} = d_u^{-\alpha} d_i^{-(1-\alpha)}$ with degrees d taken from $\hat{Y}$.

Remark 1 (Cross-fitting). In the experiments $\hat{p}$ and $\hat{Y}$ are cross-fitted over ten random folds of pairs: the nuisances used at a pair of fold k are fitted on the other nine folds (Chernozhukov et al., 2018). A nuisance therefore never depends on its own pair, which is the dependence that matters for one-hop terms. It still depends weakly on other pairs, each nuisance parameter aggregating hundreds of interactions. We measure the consequences directly by re-fitting the nuisances after interventions and attacks (Sections 6.5 and 6.7).

With $\tilde{Y} = C \odot Y$, the target is propagation over the full-exposure graph,

$$F^* = a \tilde{Y} + b T^*, \quad T^* = \tilde{Y} \tilde{Y}^\top \tilde{Y}, \quad (1)$$

with fixed constants $a, b \geq 0$. F^* is the linear LightGCN score one would compute if every item had been shown to every user. It is a normative choice: it treats potential outcomes as fixed and ignores preferences that exposure itself shapes (Chaney et al., 2018; Perdomo et al., 2020) (Section 7).

4. DRUP

Edge estimates. With clipped propensities $\bar{p} = \max(\hat{p}, \tau)$,

$$W_e^{\text{IPS}} = \frac{O_e Y_e}{\bar{p}_e}, \quad W_e^{\text{DR}} = \hat{Y}_e + \frac{O_e(Y_e - \hat{Y}_e)}{\bar{p}_e}. \quad (2)$$

Both have mean Y_e when $\bar{p} = p$.

Where the plug-in operator fails. The plug-in three-hop operator $(\tilde{W} \tilde{W}^\top \tilde{W})_{ui}$, with $\tilde{W} = C \odot W$, sums the products $\tilde{W}_{uj} \tilde{W}_{vj} \tilde{W}_{vi}$ over all walks $u-j-v-i$. If

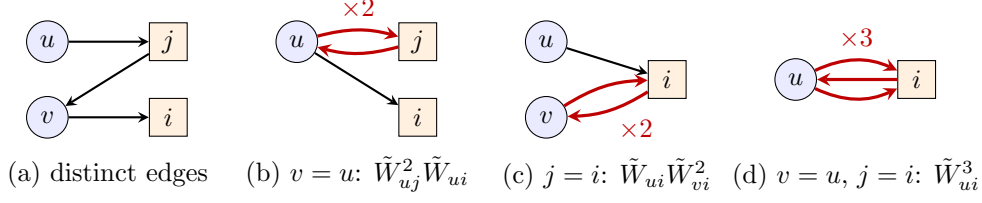


Figure 1: The four kinds of three-hop walks from user u to item i . Only walks of type (a) consist of distinct edges. In (b)–(d) an edge estimate is multiplied with itself (red), and its power has the wrong mean. DRUP replaces each repeated factor $\tilde{W}_e^k$ by $C_e^k W_e$.

$v \neq u$ and $j \neq i$ the three edges are distinct and, by independence, the product is unbiased. The other walks repeat an edge (Figure 1): for $v = u$ the edge (u, j) is traversed twice, for $j = i$ the edge (v, i) , and for $v = u, j = i$ the edge (u, i) three times. A repeated factor $\tilde{W}_e^k$ has mean $C_e^k \mathbb{E}[W_e^k] \neq C_e^k Y_e$; for IPS, $\mathbb{E}[W_e^2] = Y_e/p_e$. Because $Y_e^k = Y_e$ for binary outcomes, the unbiased replacement of $\tilde{W}_e^k$ is simply $C_e^k W_e$.

Idempotent walk correction. Let $\Delta = \tilde{W} \odot \tilde{W} - C \odot C \odot W$, with row sums r and column sums κ . The corrected three-hop operator subtracts the excess of each walk type:

$$\hat{T} = \tilde{W} \tilde{W}^\top \tilde{W} - \tilde{W} \odot (r \mathbf{1}^\top - \Delta) - \tilde{W} \odot (\mathbf{1} \kappa^\top - \Delta) - (\tilde{W}^{\odot 3} - C^{\odot 3} \odot W). \quad (3)$$

The second term removes the excess of type (b) walks, the third that of type (c), and the last that of type (d); the subtracted Δ inside the first two avoids counting type (d) twice. The DRUP score is $s = a \tilde{W} + b \hat{T}$. In the experiments $a = 1/c_1$ and $b = \beta/c_3$, where c_1 and c_3 are the mean absolute values of the one- and three-hop terms over all scored users and β is selected on validation data, so s is a fixed linear combination of the two terms.

Algorithm 1 DRUP (training-free)

Require: $\log(O, O \odot Y)$; hyper-parameters $\tau, \alpha, \lambda, \beta$

- 1: fit $\hat{p}$ (logistic exposure model) and $\hat{Y}$ (IPS-weighted additive ridge, optionally plus an IPS-weighted low-rank ALS residual), cross-fitted over ten folds of pairs
 - 2: $\bar{p} \leftarrow \max(\hat{p}, \tau)$; $W \leftarrow \hat{Y} + O \odot (Y - \hat{Y})/\bar{p}$
 - 3: $C \leftarrow d_u^{-\alpha} d_i^{-(1-\alpha)}$ with d from $\hat{Y}$
 - 4: $G \leftarrow \tilde{W}^\top \tilde{W} - \text{diag}(\tilde{W}^\top \tilde{W}) + \text{diag}(\mathbf{1}^\top (C \odot C \odot W))$ $\triangleright$ public operator
 - 5: **for** each user u (locally) **do**
 - 6: $\hat{T}_u \leftarrow \text{Eq. (4)}$; $s_u \leftarrow a \tilde{w}_u + b \hat{T}_u$; rank unexposed items by s_u
 - 7: **end for**
-

Public-operator form. Define the diagonal-corrected item Gram matrix $G = \tilde{W}^\top \tilde{W} - \text{diag}(\tilde{W}^\top \tilde{W}) + \text{diag}(\mathbf{1}^\top (C \odot C \odot W))$. Then

$$\hat{T}_u = \tilde{w}_u G - \tilde{w}_u \odot (r_u \mathbf{1}^\top - 2\delta_u) - (\tilde{w}_u^{\odot 3} - c_u^{\odot 3} \odot w_u), \quad (4)$$

where $\delta_u = \tilde{w}_u^{\odot 2} - c_u^{\odot 2} \odot w_u$ and lower-case rows belong to user u . Equation (4) matches (3) to machine precision. G is the only object shared across users (Algorithm 1). Its off-diagonal entries are products of distinct edges; its diagonal is corrected from $\sum_u \tilde{W}_{ui}^2$ to $\sum_u C_{ui}^2 W_{ui}$, the analogue of the diagonal correction of a Gram matrix estimated from masked data (Lounici, 2014).

Control-variate family. More generally, $W_e^\lambda = O_e Y_e / \bar{p}_e + \lambda(\hat{Y}_e - O_e \hat{Y}_e / \bar{p}_e)$ is unbiased for every $\lambda \in [0, 1]$ under correct propensities. It interpolates between IPS ($\lambda = 0$) and DR ($\lambda = 1$, the only member that is also robust to propensity errors), and it equals the DR estimate with the shrunk imputation

$\lambda\hat{Y}$. With $\bar{p} = p$ its variance is $\frac{1-p}{p}(Y - \lambda\hat{Y})^2$. All results below hold for every member, and λ is selected on validation data.

Imputation. Theorem 1 holds for any fixed $\hat{Y}$, so the imputation is a design choice; its accuracy enters the variance and the clipping bias (Theorem 6). We use an additive model $\mu + b_u + b_i$ fitted by IPS-weighted ridge regression, or that model plus a rank-32 residual $UV^\top$ fitted by IPS-weighted alternating least squares. The additive imputation gives every user the same item order on unexposed candidates, which makes the one-hop DR term non-personalised; the low-rank imputation removes this limitation.

Complexity and scale. One $n \times n$ Gram product and $O(mn)$ reductions, i.e., the cost of the uncorrected operator. On KuaiRec (7,176 users, 10,728 items) one configuration takes under a minute on four CPU cores. The dense W and G need $O(mn + n^2)$ memory, which limits the present implementation to catalogues of about 10^4 – 10^5 items. Because the correction only modifies diagonals and row and column sums, it carries over to sparse or low-rank representations of W , but we have not evaluated catalogues of 10^6 items.

5. Theory

Proofs are in Appendix A. Let $\varepsilon_e = |Y_e - \hat{Y}_e|$.

5.1. Unbiasedness and its limits

Theorem 1 (Edge-wise double robustness). *Under Assumptions 1–2, let $\delta_e = (p_e/\bar{p}_e - 1)(Y_e - \hat{Y}_e)$. Then $\mathbb{E}[\hat{T}_{ui}]$ equals T_{ui}^* with every distinct edge contributing $Y_e + \delta_e$ once. In particular, if every edge has a correct propensity or a correct imputation, then $\mathbb{E}[s] = F^*$.*

Recommendations rank the *unexposed* candidates of a user, and there the candidate's own edge estimate is deterministic ($W_{ui} = \hat{Y}_{ui}$ for DR, 0 for IPS). The following corollary states what DRUP estimates on the candidate set.

Corollary 2 (Unexposed candidates). *Under the conditions of Theorem 1 with $\delta \equiv 0$,*

$$\mathbb{E}[s_{ui} \mid O_{ui} = 0] = F_{ui}^* - (Y_{ui} - \hat{Y}_{ui}) g_{ui}, \quad g_{ui} = \frac{\partial F_{ui}^*}{\partial Y_{ui}} = C_{ui} \left(a + b \left(\sum_{v \neq u} C_{vi}^2 Y_{vi} + \sum_{j \neq i} C_{uj}^2 Y_{uj} + C_{ui}^2 \right) \right)$$

For IPS edges the same holds with $\hat{Y}_{ui} = 0$. The residual is the imputation error at the candidate itself, which no estimator built from the log can remove.

At three hops the walks through (u, i) are exactly those counted in g_{ui} . With IPS edges they vanish on unexposed candidates, so the uncorrected IPS operator has no repeated-walk bias there; from five hops on, walks repeat edges that are not incident to the candidate (for example $u-j-v-j-v-i$) and the bias reappears (Table 2).

Theorem 3 (Any odd number of hops). *For $K = 2r + 1$, a walk $U_0 - J_1 - U_1 - \dots - U_r - J_{r+1}$ has user variables $U_0, \dots, U_r$ and item variables $J_1, \dots, J_{r+1}$, with $U_0 = u$ and $J_{r+1} = i$. The equality pattern of an assignment is a pair π of set partitions of the user and item variables, and π fixes the multiplicity k_e of every distinct edge. Let $g_\pi(\sigma)$ be the unconstrained contraction, over the blocks of a coarser pattern $\sigma \geq \pi$, of $\prod_e C_e^{k_e} W_e$ (corrected) or of $\prod_e \tilde{W}_e^{k_e}$ (plug-in), the product running over the distinct edges of π . Then*

$$\hat{T}^{(K)} = \tilde{W}(\tilde{W}^\top \tilde{W})^r + \sum_{\pi: \exists k_e > 1} \sum_{\sigma \geq \pi} \mu(\pi, \sigma) (g_\pi^{\text{corr}}(\sigma) - g_\pi^{\text{plug}}(\sigma)),$$

with μ the Möbius function of the product of the two partition lattices, satisfies $\mathbb{E}\hat{T}^{(K)} = (\tilde{Y}\tilde{Y}^\top)^r \tilde{Y}$ under the conditions of Theorem 1 with $\delta \equiv 0$.

The number of pairs (π, σ) is 5, 99 and 2,790 for $K = 3, 5, 7$, and for $K = 3$ the construction reduces to Eq. (3). Theorem 1, Corollary 2 and Theorem 7 carry over to any odd K .

Assumption 1 excludes slates of fixed size, where exposures in a user's history are negatively dependent (Coat, for instance, collects exactly 24 ratings per user). The next result bounds the resulting bias.

Proposition 4 (Exposure dependence within users). *Suppose that exposures are independent across users but may be dependent within a user's row, with $|\text{Cov}(O_e, O_f \mid Y)| \leq \rho p_e p_f$ for distinct e, f in the same row. With correct propensities,*

$$|\mathbb{E}\hat{T}_{ui} - T_{ui}^*| \leq \rho \sum_{j \neq i} C_{uj} \left(\sum_{v \neq u} C_{vj} C_{vi} \varepsilon_{vj} \varepsilon_{vi} + C_{uj} C_{ui} \varepsilon_{uj} \varepsilon_{ui} \right).$$

For slates of k items drawn uniformly without replacement, $\rho \leq 1/k$. For DR edges the bias is of second order in the imputation error; for IPS edges ($\varepsilon = Y$) it is of first order.

Theorem 5 (Repeated-walk bias of uncorrected propagation). *Let the propensities be correct and let (u, i) be a candidate with $O_{ui} = 0$. Relative to the conditional target of Corollary 2, the uncorrected DR operator $\tilde{W}\tilde{W}^\top\tilde{W}$ has conditional bias*

$$C_{ui}\hat{Y}_{ui}(R_u^{(i)} + K_i) - C_{ui}^3\hat{Y}_{ui}(1 - \hat{Y}_{ui}^2), \quad K_i = \sum_{v \neq u} C_{vi}^2 V_{vi}, \quad R_u^{(i)} = \sum_{j \neq i} C_{uj}^2 V_{uj},$$

where $V_e = \text{Var}(W_e) = \frac{1-p_e}{p_e} \varepsilon_e^2$. Moreover: (a) $K_i \geq (1/\max_v p_{vi} - 1) \sum_{v \neq u} C_{vi}^2 \varepsilon_{vi}^2$, which is unbounded as the item's propensities vanish; (b) with clipped propensities $\bar{p} = \max(p, \tau)$, $\tau \leq 1/2$, the excess $C_e^2(\mathbb{E}[W_e^2] - \mathbb{E}[W_e])$ of a doubly traversed edge is at most $C_e^2(1/\tau - 1)\varepsilon_e^2$, with equality at $p_e = \tau$; the worst-case

repeated-walk bias is therefore $\Theta(1/\tau)$; (c) for IPS edges the conditional bias vanishes at three hops. DRUP has zero bias relative to the same target.

The within-user ranking is affected through $C_{ui}\hat{Y}_{ui}K_i$, which up-weights items whose other raters were rarely exposed; $R_u^{(i)}$ depends on i only through one term.

Theorem 6 (Variance, clipping and concentration). *Let $\Lambda_e(u, i)$ be the C -weighted mass of the walks from u to i through edge e , and $B = \max(1, 1/\tau)$. Then (a) $\text{Var}(W_e) \leq \varepsilon_e^2/\tau$; (b) $\text{Var}(\hat{T}_{ui}) \leq \frac{(1+1/\tau)^2}{\tau} \sum_e \varepsilon_e^2 \Lambda_e(u, i)^2$; (c) if $\hat{p} = p$, so that clipping is the only source of bias, $|\mathbb{E}\hat{T}_{ui} - T_{ui}^*| \leq 3\bar{\delta}(1 + \bar{\delta})^2 \sum_{j,v} C_{uj}C_{vj}C_{vi}$ with $\bar{\delta} = \max_e(1 - p_e/\tau) + \varepsilon_e$; (d) $\Pr(|\hat{T}_{ui} - \mathbb{E}\hat{T}_{ui}| \geq t) \leq 2 \exp(-2t^2 / \sum_e c_e^2)$ with $c_e = \varepsilon_e B^2 \Lambda_e(u, i)/\tau$.*

The variance scales with the imputation error ε^2 , not with Y^2 as for IPS, and τ trades variance against a clipping bias that vanishes when $\hat{Y}$ is accurate. The constants are worst-case and loose; Section 6.2 compares bound (b) with the Monte-Carlo variance.

Definition 1 (Exposure elasticity). For an intervention $do(p_i \leftarrow \pi p_i)$ on item i 's exposure mechanism, $\eta_i = \partial \log \mathbb{E}[s_{ui}] / \partial \log \pi$.

Theorem 7 (Exposure invariance). *Under Assumptions 1–2 and $\pi p \geq \tau$, DRUP has $\eta_i = 0$, also conditionally on $O_{ui} = 0$: items with equal potential outcomes receive equal expected scores whatever their exposure history. Logged-graph propagation has $\eta_i \geq 1$ (popularity amplification). Uncorrected DR propagation has $\eta_i < 0$ whenever $\hat{Y}_{ui}K_i > 0$ (over-correction).*

Exposure invariance is a statement about expected scores. It is not a distributive fairness criterion, and it is conditional: when many propensities lie below the clip, $\pi p \geq \tau$ fails and invariance is lost (Section 6.5).

5.2. Structural properties of linear propagation

The remaining results use only that, given the nuisances, s is a fixed linear function of the edge estimates. They hold for DRUP and equally for the uncorrected operators and for linear LightGCN, and they describe the recommender with its nuisances held fixed.

Proposition 8 (Affine in the user’s own data). *Fix u and a candidate i with $O_{ui} = 0$, and let $G^{(-u)}$ be G without u ’s own contribution. With frozen nuisances, $s_{ui} = \text{const} + \sum_{j \neq i} (c_{uj} G_{ji}^{(-u)} + c_{ui} w_{ui} c_{uj}^2) w_{uj}$. Hence un-logging interaction j , i.e., replacing w_{uj} by its unlogged value $\hat{Y}_{uj}$, changes s_{ui} by exactly $\phi_j = a_j(w_{uj} - \hat{Y}_{uj})$, removal effects add up over any subset, and completeness holds exactly.*

Because the removal game is additive, the ϕ_j are also its Shapley values; this is a consequence of linearity, and the attribution is relative to the stated baseline (Kumar et al., 2020).

Theorem 9 (Optimal minimal counterfactual explanation). *For candidates i, k with $s_{ui} > s_{uk}$, sorting $g_j = \phi_j^{(i)} - \phi_j^{(k)}$ in decreasing order and removing interactions until the cumulative sum exceeds the margin yields a minimum-cardinality set whose removal makes k outrank i . If the positive g_j do not sum above the margin, no such set exists. The cost is $O(n + d_u \log d_u)$.*

Theorem 10 (Certificates against injected profiles, frozen nuisances). *With frozen nuisances, one injected user v changes u ’s three-hop score of item k by*

$\Delta_k = x_k(D + a_k c_{vk}) - a_k x_k^2$, where $a = \tilde{w}_u$, $x = c_v \odot w_v$ and $D = \langle a, x \rangle$. If v logs at most L interactions with arbitrary items, labels and propensities $\geq \tau$, then D and x_k lie in closed-form intervals. Δ_k is linear in D and quadratic in x_k , so its extremes $[\underline{\Delta}_k, \overline{\Delta}_k]$ over this box are exact. Item t provably stays out of u 's top- K against any F such users if $s_{ut} + F\overline{\Delta}_t$ is below the $(K+1)$ -th largest value of $s_{uk} + F\underline{\Delta}_k$ over the other candidates. With unclipped IPS, the influence of a single interaction is unbounded.

The same argument certifies the uncorrected operators (with $\Delta_k = x_k D$). Degrees taken from $\hat{Y}$ make the fake user's weight c_v known; degrees taken from W would be attacker-controlled.

Theorem 11 (Joint differential privacy with public nuisances). *Let the population-level parts of the nuisances (the propensity model, the item parameters of $\hat{Y}$, the item degrees in C and the radius R) be computed from public data, and every user-level part from the user's own row. Scale each private row so that $\|\tilde{w}_v\|_2 \leq R$ and $\|c_v^{\odot 2} \odot w_v\|_2 \leq R^2$. Releasing G , computed over the private rows, plus symmetric Gaussian noise of scale $\sigma = \sigma_{\text{aG}}(\epsilon, \delta; 2R^2)$, the analytic Gaussian calibration (Balle and Wang, 2018), is (ϵ, δ) -DP under adding or removing one user, for every $\epsilon > 0$. Scoring by (4) uses only the release and the user's own row, so the recommender is (ϵ, δ) -jointly DP (Hsu et al., 2014). Post-processing such as low-rank denoising is free.*

Row scaling changes the estimate for users whose rows exceed R , so Theorems 1 and 7 hold for the private variant only up to this shrinkage.

Theorem 12 (Exposure-capped allocation). *Let N users each receive K of their candidates and let item i be recommended to at most cap_i users. Let $\hat{x}$*

maximise $\langle x, s \rangle$ over such allocations. The constraint matrix is totally unimodular, so the linear relaxation has an integral optimum, and min-cost flow on the scores rounded to a grid of L levels finds $\hat{x}$ exactly up to the rounding gap $\Gamma \leq NK(\max s - \min s)/L$. All caps hold, at least $NK/\max_i \text{cap}_i$ items are recommended, and for the full-exposure utility $U(x) = \langle x, F^* \rangle$,

$$U(\hat{x}) \geq \max_x U(x) - \Gamma - 2 \max_x |\langle x, s - F^* \rangle|.$$

If $\mathbb{E}s = F^*$ (DRUP with correct propensities), Theorem 6(d) and a union bound give $\max_x |\langle x, s - F^* \rangle| \leq NK t_\delta$ with probability $1 - \delta$, where $t_\delta = \max_{u,i} \left(\frac{1}{2} \sum_e c_e(u,i)^2 \ln(2|\mathcal{C}|/\delta) \right)^{1/2}$ and $\mathcal{C}$ is the candidate set. For logged-graph propagation, $\mathbb{E}s = F^*(P \odot Y)$, and the last term does not vanish.

6. Experiments

6.1. Setup

Data. *Coat* (Schnabel et al., 2016) has 290 users, 300 items and 6,960 self-selected (MNAR) ratings, exactly 24 per user. The test data are 16 ratings per user on uniformly random items (MAR); we rank only the items the user did not rate, which leaves 14.7 candidates per user on average, and 225 users have at least one positive candidate. The dataset ships propensities, which we use. *Yahoo! R3* (Marlin and Zemel, 2009) has 311,704 self-selected ratings of 15,400 users on 1,000 songs (2.0% dense) and 54,000 ratings of 10 random songs for 5,400 of the users, in the version distributed with Chen et al. (2021). A random 5% of the MAR users is held out and used only to fit an outcome-dependent Naive-Bayes propensity (Schnabel et al., 2016); in a preliminary comparison on validation data it did not improve the propagation

operators, and all reported results use the logistic exposure model. The remaining 5,130 test users have 10 candidates each, and 2,315 of them have a positive. *KuaiRec* (Gao et al., 2022) provides the big matrix (7,176 users, 10,728 items, 10.3M logged views, 13.4% dense) as the MNAR log and a fully observed small matrix (1,411 users, 3,327 items) as the test set. We rank all small-matrix items that the user never viewed in the log (3,314 candidates per user, 154 of them positive on average). Positives are ratings ≥ 4 (Coat, Yahoo!R3) and a watch ratio ≥ 2 (KuaiRec). A random ranking reaches nDCG@5 of 0.306 on Coat and 0.352 on Yahoo!R3 and nDCG@20 of 0.046 on KuaiRec. The candidate sets of Coat and Yahoo!R3 are small, so their metric differences are compressed; KuaiRec ranks thousands of candidates.

Protocol. The unbiased data are split into a validation part (30% of each user’s candidates on Coat, 30% of the test users otherwise) and a test part, over 10 (Coat) or 5 random splits. Every method selects its configuration on the validation part of each split, and trained models also select their epoch there. All fitted nuisances are cross-fitted over ten folds of pairs (Remark 1). We report the mean and standard deviation over splits of the test metric. Because paired tests over re-sampled splits are anti-conservative when the splits overlap (Nadeau and Bengio, 2003), differences are tested at the level of users, with a Holm correction over all comparisons of a dataset (Appendix B). Progress claims in top- N recommendation have repeatedly failed against well-tuned simple baselines (Ferrari Dacrema et al., 2019; Rendle et al., 2019; Anelli et al., 2023); we therefore tune every method on the same validation data and report how many configurations each one searched.

Methods. We compare 23 methods. *Non-personalised:* popularity and the imputation $\hat{Y}$ alone. *Trained with a pointwise loss:* MF, IPS-MF (Schnabel et al., 2016), DR-MF (Wang et al., 2019b), LightGCN and DR-LightGCN (LightGCN with the DR loss). *Trained with a pairwise (BPR) loss:* MF, PDA (Zhang et al., 2021), LightGCN (He et al., 2020), r-AdjNorm (Zhao et al., 2022) and NAVIP (Kim et al., 2022). *Training-free on the logged graph:* linear LightGCN (the one- plus three-hop form of Section 4 on $O \odot Y$; with the searched normalisation exponent it is a linear r-AdjNorm), EASE (Steck, 2019) and GF-CF (Shen et al., 2021). *Training-free on a debiased graph:* IPS adjacency (NAVIP-style), IPS adjacency with the walk correction, EASE and GF-CF on the DR graph W , DR adjacency, DRUP, and the five-hop versions of DR adjacency and DRUP. Grids, the number of configurations per method, and training details are in Appendix B. The training-free methods search between 3 and 6,300 configurations and the trained ones between 4 and 36; each trained configuration is trained once and evaluated after every epoch on every split, so that it is also tuned per split.

6.2. Monte-Carlo checks

Table 2 uses a synthetic MNAR model with known propensities: 30 users, 40 items, low-rank preferences, and exposures driven by user activity, item popularity and the outcome itself (median propensity 0.06, a third of the pairs below 0.05). We draw 20,000 exposure matrices for three hops and 4,000 for five hops.

Over all entries, the uncorrected IPS and DR operators are biased by 2.3 and 1.8 times the mean target at three hops, and by 15 and 8 times at five hops. With the correction the residual bias (0.02–0.06) is at the level

Table 2: Monte-Carlo check of Theorems 1, 3, 5 and 7 and Corollary 2. Relative $|\text{bias}|$ is the mean absolute bias divided by the mean target, over all entries and conditionally on unexposed candidates (against the conditional target of Corollary 2). Kendall τ is the mean within-user rank agreement of one realisation with the full-exposure target on unexposed candidates. η is the exposure elasticity (separate model, 3,000 replications).

Estimator	rel. $ \text{bias} $, $K = 3$		rel. $ \text{bias} $, $K = 5$		Kendall τ	elasticity
	all	unexposed	all	unexposed	($K = 3$)	η
Logged graph	–	–	–	–	–	+1.777
IPS adjacency	2.335	0.029	14.641	4.667	0.024	+0.007
IPS + walk correction	0.028	0.029	0.060	0.059	0.024	+0.007
DR adjacency	1.756	1.137	8.350	5.053	0.199	-0.187
DRUP	0.019	0.019	0.039	0.040	0.122	+0.005

of the Monte-Carlo error. On unexposed candidates the picture is the one predicted by Corollary 2: the uncorrected IPS operator is unbiased at three hops (0.029, identical to IPS with the correction) but not at five hops (4.7), and the uncorrected DR operator is biased at both depths (1.1 and 5.1). The exposure elasticity is +1.78 for the logged graph, -0.19 for DR adjacency and +0.005 for DRUP. The K -hop engine reproduces the closed-form three-hop operator to 7×10^{-15} and brute-force enumeration of all five-hop walks to 10^{-13} .

Unbiasedness does not buy ranking accuracy in this model. The rank agreement of a single realisation with the target is higher for the uncorrected DR operator (Kendall $\tau = 0.20$) than for DRUP (0.12), and at five hops 0.11 against 0.09. The reason is visible in Theorem 5: on unexposed candidates the bias is $C_{ui}\hat{Y}_{ui}(R_u^{(i)} + K_i)$, which adds a positive multiple of the imputation to the score, and here the imputation is informative. The corrected operator

removes this term and, with it, some useful signal. For IPS edges, whose bias does not involve the imputation, the correction leaves the rank agreement unchanged (0.061 against 0.059 at five hops).

Stress tests outside the assumptions of Theorem 1 are in the Supplementary Material (Table S1). With fixed-size slates, which make exposures within a user negatively dependent, IPS picks up a first-order bias (0.22) whereas DRUP stays at 0.024, as Proposition 4 predicts; with misspecified propensities DRUP is exactly unbiased when the imputation is exact and biased by 0.16 when it is noisy, against 0.57 for IPS and 1.9 for the uncorrected DR operator. The bounds of Theorem 6 were never violated but are loose by a factor of 70 to 8,000 (Table S2).

6.3. Accuracy

Tables 3 and 4 report accuracy and the user-level tests.

The walk correction does not change accuracy. On every dataset DRUP and the uncorrected DR adjacency with the same grid are statistically indistinguishable (Coat +0.0014, KuaiRec −0.0001 with $p_{\text{Holm}} = 0.45$), and on Yahoo! R3 they coincide exactly, because validation selects the IPS end of the control-variate family ($\lambda = 0$) on every split and IPS edges have no three-hop repeated-walk bias on unexposed candidates (Corollary 2). The IPS adjacency and IPS with the correction give identical rankings on all three datasets for the same reason. At five hops on KuaiRec, where the correction is active for both edge types, the corrected operator is lower by 0.0003 nDCG@20 ($p_{\text{Holm}} = 0.04$ with the t -test, $p = 0.66$ with the Wilcoxon test), a negligible difference in line with the ranking agreement of Section 6.2. Five

hops themselves do not add accuracy over three (0.6376 against 0.6378).

Propagation over the DR graph is what helps on KuaiRec. On KuaiRec, whose test matrix is fully observed and whose candidate sets have 3,314 items, training-free propagation over the DR graph reaches nDCG@20 of 0.638 and is significantly better than every trained model (the best, LightGCN with a pointwise loss, reaches 0.628; $+0.010$, 95% CI $[0.007, 0.013]$), every operator on the logged graph (GF-CF 0.614, IPS adjacency 0.613, linear LightGCN 0.604) and the low-rank imputation used alone (0.619). GF-CF applied to the same DR graph performs as well as DRUP (0.637), which confirms that the gain comes from the DR graph and not from a particular filter. The selected DR configurations use the low-rank imputation on all five splits.

Yahoo!R3 and Coat. On Yahoo!R3 the log is 2% dense and the imputation alone is below popularity (0.516 against 0.533), so almost every edge of a DR graph would be imputed from a poor model; validation therefore selects IPS edges ($\lambda = 0$) with cross-fitted W -degrees on every split. With these settings DRUP (0.657) is on par with linear propagation over the logged graph (0.659) and GF-CF (0.657), and it is significantly behind the three graph models trained with BPR: r-AdjNorm (0.671; -0.015 , $p_{\text{Holm}} = 0.004$), NAVIP and LightGCN (0.669; $p_{\text{Holm}} \leq 0.04$). With degrees taken from the imputation instead, DRUP drops to 0.62–0.63 on this dataset, so the degree source matters here. On Coat (290 users, 15 candidates per user) DRUP is significantly better than pointwise MF and IPS-MF, and no other difference survives the Holm correction except GF-CF on the DR graph, which is better by 0.013 ($p_{\text{Holm}} = 0.03$); the best models are again the BPR graph models

(r-AdjNorm 0.575, LightGCN 0.570), but their advantage over DRUP is not significant at the user level.

Baselines. Two changes of the baselines matter. First, pairwise training is decisive on Yahoo! R3: pointwise MF variants stay near 0.51, below popularity, whereas BPR-MF reaches 0.646 and BPR-trained LightGCN 0.669. On KuaiRec the ordering reverses: pointwise LightGCN (0.628) and MF (0.623) beat their BPR versions (0.609, 0.591). Second, per-split selection of configuration and epoch benefits the trained models on the small Coat data, where re-training the split-0 selection with other seeds lowers LightGCN from 0.581 to 0.547 ± 0.018 ; on KuaiRec and for the BPR graph models on Yahoo! R3 the seed standard deviation is below 0.005 (at most 0.010 for the other Yahoo! R3 models). Across the three datasets, the best trained model is thus a BPR graph model on Coat and Yahoo! R3 and a pointwise one on KuaiRec, while DR propagation without training is the most accurate method on the largest and most complete test set.

6.4. Does the correction matter on sparser logs?

Theorem 5 predicts a larger bias when propensities are small. We therefore thinned the KuaiRec log to 10% and 3% of its exposures, which lowers the median logged propensity from 0.48 to 0.045 and 0.013, re-fitted the nuisances and re-selected every operator (Table 5). The corrected and the uncorrected operators still rank almost identically: their top-20 lists overlap by at least 93%, the rank correlation is at least 0.999, and the accuracy differences stay below 0.0005 nDCG@20 in either direction, at three and at five hops. On these data the repeated-walk term is small compared with the

score differences that decide the ranking, and the value of the correction lies in the properties of the scores, not in the rankings they induce.

6.5. Exposure invariance on real data

Table 6 tests Theorem 7 with a known intervention on exposure. The logged graph behaves as the theorem predicts for correlational propagation: with frozen nuisances $\eta = +1.00$ to $+1.03$ on all three datasets, and thinned items drop by 0.07–0.24 in mean within-user rank. On KuaiRec and Coat, DRUP is invariant to within a few hundredths ($\eta = -0.012$ and -0.008) and stays so when its nuisances are re-fitted on the thinned log, with the known or with re-estimated propensities where they are estimated ($|\eta| \leq 0.022$). IPS adjacency is far from invariant ($\eta = +0.14$ on KuaiRec, $+0.90$ on Coat). The uncorrected DR adjacency is as invariant as DRUP on these data; its over-correction ($\eta = -0.19$ in Table 2) scales with $(1-p)/p \cdot \varepsilon^2$ (Theorem 5), and the selected clips and imputations keep it small here.

Yahoo! R3 is the exception, and Table 7 explains it. There, validation selects IPS edges ($\lambda = 0$) with a clip of 0.1–0.2, which binds for 82% of the logged pairs at $\tau = 0.2$. A clipped IPS edge is $O_e Y_e / \tau$, proportional to the exposure itself, so DRUP inherits the elasticity of the logged graph ($\eta = +0.96$). On Coat the clip binds just as often (77%), but with DR edges ($\lambda = 1$) and an accurate imputation the clipping bias is small (Theorem 6(c)) and invariance survives. Lowering the clip on Yahoo! R3 reduces the elasticity to $+0.18 \pm 0.11$ at $\tau = 5 \cdot 10^{-4}$, where almost no propensity is clipped, at a cost of 0.044 nDCG@5; the remaining elasticity and its large spread over the thinning draws come from heavy-tailed inverse-propensity weights on a log that is 2% dense. Exposure invariance is therefore a property of the

estimator under its assumptions, and whether a tuned configuration has it must be checked; the thinning audit is a simple way to do so.

Table 8 shows the distributional side. On KuaiRec, the exposure-conditional bias drops from 0.83 (logged graph) and 0.80 (IPS) to 0.33 for DRUP and DR adjacency: given their true quality, items are ranked much less by how often they were shown. At the same time DRUP’s top-20 lists are concentrated on 1.2% of the candidate items, against 15% for the logged graph, because the estimated full-exposure scores are themselves concentrated on few items; on Coat its Gini coefficient is 0.60 against 0.34. Exposure invariance and exposure concentration are different properties, and DRUP improves the first at the expense of the second. The user-group gaps do not differ between operators. They are clearly non-zero on Coat (men 0.11 above women) and on Yahoo! R3 (inactive users 0.046 below active ones), under every operator, and not on KuaiRec.

6.6. *Exposure-capped allocation*

Table 9 applies the exposure-capped allocation of Theorem 12 to every operator. The min-cost-flow solver returns the optimum up to its rounding gap (a relative gap below $4 \cdot 10^{-5}$ in every run), all caps hold, and coverage never falls below the guaranteed $NK/\text{max cap}$. On KuaiRec, where exposure is extremely concentrated, DR adjacency and DRUP are the most accurate operators under every cap, for example 0.199 against 0.173 (logged graph) and 0.178 (IPS) at $c = 5$, and 0.140 against 0.115 and 0.121 at $c = 2$. Under the same cap, however, their lists remain more concentrated (Gini 0.80 against 0.72 at $c = 5$), and at equal Gini the frontier over each operator’s grid (Table S4) favours the correlation-based operators on Coat. On Coat, whose

random test items are already diverse, capping costs DRUP more accuracy than it costs the logged graph; on Yahoo!R3 exposure is not concentrated (Gini 0.27) and caps down to twice the fair share cost every operator at most 0.002. The allocation is thus the mechanism that controls concentration, and the DR graph is the better input to it when exposure in the log is very uneven.

6.7. Structural properties

The properties of Section 5.2 hold for the model with frozen (explanations, certificates) or public (privacy) nuisances, and for any linear operator; we check them for DRUP and report where they are informative.

Explanations. Attributions recomputed against full re-propagation are exact to floating-point precision (relative error 10^{-16} in float64 on Coat and Yahoo!R3, $3 \cdot 10^{-6}$ in float32 on KuaiRec; Table 10), and removing the interactions with the largest attribution moves the top item further down than removing random or similar ones, so the attributions are faithful. The counterfactual analysis mostly certifies that no explanation from the user’s own data exists: on Coat 94% of the top recommendations cannot be overturned by removing any subset of the user’s own interactions, on KuaiRec 52%, and on Yahoo!R3 10%; the minimal explanation otherwise has a median size of 2, 7 and 1 interactions. Such recommendations rest on population-level evidence (through G), which the attribution to a user’s own interactions does not explain.

Robustness. No certificate was violated in any run (Table 11). With imputation-based degrees, DR adjacency and DRUP certify every Coat user against one

or two fake profiles and 93–95% of users on Coat and KuaiRec against five, and realised attacks with up to 100 fake users never placed the target in a KuaiRec top-20. Certificates are informative only for small budgets: at $F = 20$ they cover 10–26% of users, and at 200 fake users the target reaches 39% of KuaiRec users under DR and DRUP, more than under IPS (10%) and the logged graph (29%). On Yahoo!R3, where DRUP selects IPS edges with cross-fitted W -degrees, the fake users’ degrees are attacker-controlled and no user is certified. When the nuisances are re-fitted on the poisoned log, which the certificates do not cover, none of the certified user–target pairs of DRUP (5,196 on Coat, 2,821 on KuaiRec) was violated, and the realised hit rates were lower than with frozen nuisances (Table S5 in the Supplementary Material), because the fake users’ own imputation raises their degree and lowers their weight.

Privacy. Table 12 reports the jointly private variant of Theorem 11, in which only public users inform the population-level nuisances. Useful privacy needs a large user base and a weak guarantee. On KuaiRec, private DRUP reaches nDCG@20 of 0.596 at $\epsilon = 16$ and 0.530 at $\epsilon = 8$, against 0.637 without privacy and 0.577 for popularity, and it is below popularity for $\epsilon \leq 8$. On Yahoo!R3 it reaches 0.574 at $\epsilon = 16$ (non-private 0.652, popularity 0.525). On Coat the public-nuisance model itself is weak, because 29 public users cannot estimate the item parameters: without any noise it reaches only 0.454, below popularity. These numbers are lower than a variant that treats all nuisances as public would suggest, and they show the cost of an honest privacy accounting; privatising the nuisance models with a composed budget is left open.

7. Discussion

What the correction is for. On the three benchmarks the walk correction does not change top- K accuracy, and in simulation the uncorrected DR operator can even agree slightly better with the full-exposure ranking, because its bias adds weight to the imputation (Section 6.2). The correction matters wherever the *value* of a propagated score is used rather than only its order within one user’s list: when scores are compared across users, as in the exposure-capped allocation; when they are aggregated into an estimate of full-exposure utility; and when an auditor wants to verify that expected scores do not depend on exposure. In these uses the uncorrected estimate is biased by 1.8 to 15 times the target in our simulations, and the bias grows with the number of hops and with small propensities. A practitioner who only needs within-user rankings from a dense log loses nothing by using the uncorrected DR adjacency; one who needs calibrated, auditable scores, or propagates over five hops or more, should use the corrected operator. In both cases, most of the accuracy comes from training-free propagation over a doubly robust graph.

Stakeholders. For *item providers*, exposure invariance means that an item is not penalised in expectation for having been shown rarely in the past. It does not mean equal exposure: when quality is concentrated, DRUP’s lists can be more concentrated than those of the logged graph (Table 8), and the scores of rarely exposed items have a larger variance (Theorem 6(a)), so their realised rankings are noisier. Providers who need exposure guarantees need an explicit mechanism such as the exposure-capped allocation. For *users*, the attributions show which of their own interactions drive a recommendation, or certify that none can overturn it; the population-level part of the

score remains unexplained (it decides 94% of the top recommendations on Coat and 52% on KuaiRec), and explaining it through G would conflict with the privacy guarantee. For the *platform*, DRUP is cheap and training-free but needs propensities, and the clip τ and the control-variate weight λ trade exposure invariance against accuracy. This trade-off is a policy choice rather than a tuning detail. One option is to select the most accurate configuration subject to a bound on the elasticity measured in a thinning audit such as ours. For *auditors and regulators*, DRUP produces concrete artefacts: a single item operator, a thinning audit with a known answer, and certificates. Under the EU Digital Services Act (European Parliament and Council of the European Union, 2022), platforms must explain the main parameters of their recommender systems (Art. 27), assess and mitigate systemic risks (Arts. 34–35), undergo independent audits (Art. 37) and give vetted researchers access to data (Art. 40). Our artefacts fit the last three, but the audit of exposure invariance needs the platform’s propensities, which an external auditor usually does not have.

Limitations. The analysis is static. It assumes a single logging policy and fixed potential outcomes, whereas recommenders are retrained on the feedback they generate (Chaney et al., 2018; Hazrati and Ricci, 2022) and can change the preferences they predict (Perdomo et al., 2020); extending the guarantees to adaptive and slate logging (Swaminathan et al., 2017) is open. Proposition 4 covers dependence within a user’s history, not dependence across users. The idempotent correction relies on binary outcomes; graded relevance would need separate unbiased estimates of the powers of Y . Items that were never exposed fall outside $\pi p \geq \tau$ and are scored by the imputa-

tion only. The nuisances are cross-fitted rather than fitted on an independent log, the joint-DP guarantee requires public nuisances, and on the small Coat data the public-nuisance variant loses much of its accuracy. The certificates are informative only against a handful of fake profiles (on KuaiRec, 95% of users at five profiles, none at fifty), and the variance and concentration bounds are loose by several orders of magnitude (Section 6.2). The target F^* is a linear LightGCN score of the full-exposure world, a normative choice that other stakeholders may not share. Like every method in our comparison, DRUP selects its hyper-parameters on a small unbiased sample; without such a sample, selection would need an off-policy estimate of the validation metric. Finally, our largest catalogue has about 10^4 items.

8. Implications

For research. Propagation-level debiasing needs to consider the propagation operator itself, not only the edges it propagates: multiplying unbiased edge estimates does not give an unbiased multi-hop score. The repeated-walk bias is exactly removable, and the tools that remove it (idempotent reduction and Möbius inversion over walk-index coincidences) apply to any polynomial graph filter built from inverse-propensity or DR estimates. Our evaluation also shows that training-free propagation over a DR graph is a strong baseline against which trained debiased graph recommenders should be compared, and that claims of exposure invariance can be tested on real data by thinning exposures with a known factor.

For practice. DRUP runs at the cost of one item Gram matrix and needs no gradient training, so it can serve as an auditable baseline or as a candidate

scorer. The choice between the corrected and the uncorrected operator depends on whether score values or only within-user orders matter. On sparse logs, exact exposure invariance requires small clips and costs accuracy (Table 7). The structural properties (exact attributions, certificates, a private item operator and capped allocation) come with any linear propagation operator and hold for the model with its nuisances held fixed; platforms should report them together with that scope.

9. Conclusion

Inverse-propensity and doubly robust graph propagation are biased beyond one hop because walks that repeat an edge multiply its estimate with itself. We gave the exact bias, whose worst case grows as the inverse of the propensity clip, and removed it with an idempotent correction that is exact for any odd number of hops and costs no more than the uncorrected operator. With fixed nuisances the resulting estimator, DRUP, is edge-wise doubly robust for propagation over the full-exposure graph, robust to within-user exposure dependence to second order in the imputation error, and exposure-invariant in expectation. On three benchmarks with unbiased test data, training-free DR propagation is competitive with tuned trained and closed-form baselines, while the correction itself leaves top- K accuracy unchanged. Future work includes private nuisance models, certificates that remain informative for large catalogues, guarantees under adaptive logging, and combining the correction with learned propagation weights.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used generative AI tools to support code development and the development of ideas for the paper. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Appendix A. Proofs

Throughout, expectations are conditional on Y , and “walk” means a three-hop walk $u - j - v - i$ unless stated otherwise. Its edges are (u, j) , (v, j) and (v, i) .

Equations (3) and (4). Split the plug-in sum $(\tilde{W}\tilde{W}^\top\tilde{W})_{ui} = \sum_{j,v} \tilde{W}_{uj}\tilde{W}_{vj}\tilde{W}_{vi}$ by walk type (Figure 1): (a) $v \neq u, j \neq i$; (b) $v = u, j \neq i$, giving $\sum_{j \neq i} \tilde{W}_{uj}^2 \tilde{W}_{ui}$; (c) $v \neq u, j = i$, giving $\sum_{v \neq u} \tilde{W}_{ui} \tilde{W}_{vi}^2$; (d) $v = u, j = i$, giving $\tilde{W}_{ui}^3$. The corrected estimator keeps (a) and replaces (b)–(d) by $\sum_{j \neq i} C_{uj}^2 W_{uj} \tilde{W}_{ui}$, $\sum_{v \neq u} \tilde{W}_{ui} C_{vi}^2 W_{vi}$ and $C_{ui}^3 W_{ui}$. The difference between plug-in and corrected sums is $\tilde{W}_{ui} \sum_{j \neq i} \Delta_{uj} + \tilde{W}_{ui} \sum_{v \neq u} \Delta_{vi} + (\tilde{W}_{ui}^3 - C_{ui}^3 W_{ui}) = \tilde{W}_{ui}(r_u - \Delta_{ui}) + \tilde{W}_{ui}(\kappa_i - \Delta_{ui}) + (\tilde{W}_{ui}^3 - C_{ui}^3 W_{ui})$, which is Eq. (3). For Eq. (4), $(\tilde{w}_u G)_i = \sum_{j \neq i} \tilde{W}_{uj}(\tilde{W}^\top \tilde{W})_{ji} + \tilde{W}_{ui} \sum_v C_{vi}^2 W_{vi} = (\tilde{W}\tilde{W}^\top\tilde{W})_{ui} -$

$\tilde{W}_{ui}\kappa_i + \tilde{W}_{ui} \sum_v C_{vi}^2 W_{vi}$, and substituting into Eq. (3) cancels the κ_i terms. $\square$

Theorem 1.. By the decomposition above, the corrected estimator is a sum over walks of $\prod_e C_e^{k_e} W_e$, the product running over the *distinct* edges of the walk with multiplicities k_e . Under Assumptions 1–2 the W_e of distinct edges are independent, so the expectation factorises, with $\mathbb{E}W_e = \hat{Y}_e + p_e(Y_e - \hat{Y}_e)/\bar{p}_e = Y_e + \delta_e$. The target walk monomial is $\prod_e C_e^{k_e} Y_e^{k_e} = \prod_e C_e^{k_e} Y_e$ because $Y_e \in \{0, 1\}$. The one-hop term is unbiased edge by edge. $\square$

Corollary 2.. Given $O_{ui} = 0$, $W_{ui} = \hat{Y}_{ui}$ is deterministic and the other edge estimates keep their distribution (Assumption 1). The edge (u, i) belongs to the one-hop term and to exactly the walks of types (b), (c) and (d). In each of them it contributes $C_{ui}^k \hat{Y}_{ui}$ instead of $C_{ui}^k Y_{ui}$, and every other distinct edge contributes its mean Y_e . After the idempotent reduction, F_{ui}^* is affine in Y_{ui} , so $\mathbb{E}[s_{ui} \mid O_{ui} = 0] = F_{ui}^*|_{Y_{ui} \leftarrow \hat{Y}_{ui}} = F_{ui}^* - (Y_{ui} - \hat{Y}_{ui}) \partial F_{ui}^* / \partial Y_{ui}$. The derivative collects aC_{ui} from the one-hop term, $bC_{ui} \sum_{j \neq i} C_{uj}^2 Y_{uj}$ from type (b), $bC_{ui} \sum_{v \neq u} C_{vi}^2 Y_{vi}$ from type (c) and bC_{ui}^3 from type (d). For IPS edges, $W_{ui} = 0$ given $O_{ui} = 0$. $\square$

Theorem 3.. Fix u and i . For an assignment φ of values to the free variables, let $\pi(\varphi)$ be its equality pattern. Two walk edges (U_s, J_t) and $(U_{s'}, J_{t'})$ coincide if and only if $U_s, U_{s'}$ share a user block and $J_t, J_{t'}$ share an item block, so the distinct edges and their multiplicities k_e depend on $\pi(\varphi)$ only. For a pattern π , let $F_\pi(\varphi) = \prod_\epsilon X_{\varphi(\epsilon)}^{(k_\epsilon)}$, the product over the distinct edges ϵ of π , with $X^{(k)} = C^{\odot k} \odot W$ (corrected) or $X^{(k)} = \tilde{W}^{\odot k}$ (plug-in). The plug-in K -hop product equals $\sum_\pi S_\pi^{\text{plug}}$ and the corrected estimator is $\sum_\pi S_\pi^{\text{corr}}$, where $S_\pi =$

$\sum_{\varphi:\pi(\varphi)=\pi} F_{\pi}(\varphi)$. The two coincide for patterns without a repeated edge, which gives the form of $\hat{T}^{(K)}$ once each S_{π} is written through unconstrained sums. For $\sigma \geq \pi$ let $g_{\pi}(\sigma) = \sum_{\varphi:\pi(\varphi)\geq\sigma} F_{\pi}(\varphi)$: variables in the same block of σ are equal and values of different blocks are unrestricted, so $g_{\pi}(\sigma)$ is a single tensor contraction over the blocks of σ (with the blocks of U_0 and J_{r+1} fixed to u and i). Grouping the terms of $g_{\pi}(\sigma)$ by their exact pattern gives $g_{\pi}(\sigma) = \sum_{\tau \geq \sigma} f_{\pi}(\tau)$ with $f_{\pi}(\tau) = \sum_{\varphi:\pi(\varphi)=\tau} F_{\pi}(\varphi)$. Möbius inversion on the interval $[\pi, \hat{1}]$ of the product of the two partition lattices (Rota, 1964) yields $S_{\pi} = f_{\pi}(\pi) = \sum_{\sigma \geq \pi} \mu(\pi, \sigma) g_{\pi}(\sigma)$, where $\mu(\pi, \sigma) = \prod_B (-1)^{b_B-1} (b_B - 1)!$ over the blocks B of σ in both lattices and b_B is the number of blocks of π merged into B . Finally, for every φ with $\pi(\varphi) = \pi$ the distinct edges are independent, so $\mathbb{E}F_{\pi}^{\text{corr}}(\varphi) = \prod_{\epsilon} C_{\varphi(\epsilon)}^{k_{\epsilon}} Y_{\varphi(\epsilon)}$, which is the corresponding monomial of $(\tilde{Y}\tilde{Y}^{\top})^r \tilde{Y}$ since Y is binary. The number of contraction terms is the number of pairs (π, σ) with $\sigma \geq \pi$ and π containing a repeated edge, which is 5, 99 and 2,790 for $K = 3, 5, 7$. $\square$

Proposition 4. With correct propensities $\mathbb{E}W_e = Y_e$, and for distinct e, f in the same row $\text{Cov}(W_e, W_f) = (Y_e - \hat{Y}_e)(Y_f - \hat{Y}_f) \text{Cov}(O_e, O_f)/(p_e p_f)$, so $|\text{Cov}(W_e, W_f)| \leq \rho \varepsilon_e \varepsilon_f$. In a type (a) walk the edge (u, j) lies in row u and the edges $(v, j), (v, i)$ in row $v \neq u$; rows are independent, so $\mathbb{E}[W_{uj}W_{vj}W_{vi}] = Y_{uj}(Y_{vj}Y_{vi} + \text{Cov}(W_{vj}, W_{vi}))$ and the bias of the walk is at most $C_{uj}C_{vj}C_{vi}\rho \varepsilon_{vj}\varepsilon_{vi}$. A corrected type (b) walk contributes $C_{uj}^2 C_{ui} W_{uj} W_{ui}$ with two edges in row u , with bias at most $C_{uj}^2 C_{ui} \rho \varepsilon_{uj} \varepsilon_{ui}$. Type (c) walks have their two distinct edges in different rows and type (d) walks a single edge, so they are unbiased. Summing gives the bound. For k of n items drawn uniformly, $\Pr(O_e = O_f = 1) = k(k-1)/(n(n-1))$, so $|\text{Cov}(O_e, O_f)|/(p_e p_f) = (n-k)/(k(n-1)) \leq 1/k$. $\square$

Theorem 5. By the decomposition of Eq. (3), the plug-in minus the corrected operator is $\tilde{W}_{ui} \sum_{j \neq i} \Delta_{uj} + \tilde{W}_{ui} \sum_{v \neq u} \Delta_{vi} + (\tilde{W}_{ui}^3 - C_{ui}^3 W_{ui})$. Given $O_{ui} = 0$, $\tilde{W}_{ui} = C_{ui} \hat{Y}_{ui}$ is deterministic and the other Δ_e keep their distribution. With correct propensities and binary Y , $\mathbb{E}\Delta_e = C_e^2(\mathbb{E}W_e^2 - \mathbb{E}W_e) = C_e^2(\text{Var } W_e + Y_e^2 - Y_e) = C_e^2 V_e$ with $V_e = \frac{1-p_e}{p_e} \varepsilon_e^2$, which gives the stated bias, the last term being $C_{ui}^3(\hat{Y}_{ui}^3 - \hat{Y}_{ui})$. Part (a) uses $(1-p)/p \geq 1/\max_v p_{vi} - 1$. For (b), with $\mu_e = \mathbb{E}W_e \in [0, 1]$, $\mathbb{E}W_e^2 - \mathbb{E}W_e = \text{Var } W_e - \mu_e(1 - \mu_e) \leq \text{Var } W_e = p_e(1-p_e)\varepsilon_e^2/\bar{p}_e^2$. If $p_e \geq \tau$ this is $\frac{1-p_e}{p_e} \varepsilon_e^2 \leq (1/\tau - 1)\varepsilon_e^2$; if $p_e < \tau \leq 1/2$ it is $p_e(1-p_e)\varepsilon_e^2/\tau^2 < \tau(1-\tau)\varepsilon_e^2/\tau^2$ because $p(1-p)$ increases on $[0, 1/2]$. At $p_e = \tau$, $\mu_e = Y_e$ and the bound is attained; setting all $p_{vi} = \tau$ gives a bias $C_{ui} \hat{Y}_{ui}(1/\tau - 1) \sum_{v \neq u} C_{vi}^2 \varepsilon_{vi}^2$, so the worst case is $\Theta(1/\tau)$. For (c), IPS gives $\tilde{W}_{ui} = 0$ and $C_{ui}^3 W_{ui} = 0$ given $O_{ui} = 0$. $\square$

Theorem 6. (a) $\text{Var}(W_e) = p_e(1-p_e)\varepsilon_e^2/\bar{p}_e^2$; if $p_e \geq \tau$ it is at most $(1/\tau - 1)\varepsilon_e^2$, otherwise at most $p_e\varepsilon_e^2/\tau^2 < \varepsilon_e^2/\tau$. (b) $\hat{T}_{ui}$ is multilinear in the independent W_e and hence affine in each of them. The Efron–Stein inequality (Efron and Stein, 1981) gives $\text{Var}(\hat{T}) \leq \sum_e \mathbb{E}[(\partial_e \hat{T})^2] \text{Var}(W_e)$. The derivative $\partial_e \hat{T}$ is a sum over the walks through e of C -weights times at most two other independent W_f , and $\mathbb{E}W_f^2 = \mu_f^2 + \text{Var } W_f \leq 1 + 1/\tau$. Minkowski’s inequality bounds $\|\partial_e \hat{T}\|_2 \leq (1 + 1/\tau)\Lambda_e$. (c) By Theorem 1, the bias is at most $\sum_{\text{walks}} |\kappa_{\text{walk}}| [\prod_f (Y_f + |\delta_f|) - \prod_f Y_f] \leq ((1 + \bar{\delta})^3 - 1) \sum_{j,v} C_{uj} C_{vj} C_{vi}$, and $(1+x)^3 - 1 \leq 3x(1+x)^2$. With $\hat{p} = p$, $|\delta_e| = (1 - p_e/\tau)_+ \varepsilon_e$. (d) Changing O_e moves W_e by at most ε_e/τ . Every edge estimate satisfies $|W_f| \leq B$, so $|\partial_e \hat{T}| \leq B^2 \Lambda_e$, and McDiarmid’s inequality (McDiarmid, 1989) applies. $\square$

Theorem 7. If $\pi p \geq \tau$ the clip does not bind after the intervention, so the propensities remain correct. DRUP: by Theorem 1, $\mathbb{E}[s] = F^*$, and by

Corollary 2, $\mathbb{E}[s \mid O_{ui} = 0]$ is F^* with Y_{ui} replaced by the fixed $\hat{Y}_{ui}$; neither depends on π . Logged graph: $\mathbb{E}[O_e Y_e] = p_e Y_e$, and every walk that ends at i contains $k \geq 1$ edges incident to i , so its expectation scales as π^k and η_i is an average of such k . Uncorrected DR: by Theorem 5, $\mathbb{E}[s_{ui} \mid O_{ui} = 0]$ equals the DRUP target plus $C_{ui} \hat{Y}_{ui} (R_u^{(i)} + K_i(\pi))$ minus a term free of π ; $R_u^{(i)}$ involves no edge of item i , and $K_i(\pi) = \sum_{v \neq u} C_{vi}^2 \frac{1 - \pi p_{vi}}{\pi p_{vi}} \varepsilon_{vi}^2$ is strictly decreasing in π . $\square$

Proposition 8 and Theorem 9.. With frozen nuisances, substitute $G = G^{(-u)} + \tilde{w}_u^\top \tilde{w}_u - \text{diag}(\tilde{w}_u^2) + \text{diag}(c_u^2 w_u)$ into (4). The terms $\tilde{w}_{ui} \sum_j \tilde{w}_{uj}^2$ cancel against the local correction, leaving $\tilde{w}_{ui} \sum_j c_{uj}^2 w_{uj}$, and every remaining term involving w_{uj} ($j \neq i$) is linear. Un-logging j replaces w_{uj} by $\hat{Y}_{uj}$, which changes s_{ui} by $a_j(w_{uj} - \hat{Y}_{uj})$; the game is additive, so its Shapley values are the individual effects. For Theorem 9, the margin after removing S is $s_{ui} - s_{uk} - \sum_{j \in S} g_j$, so minimising $|S|$ under $\sum_S g_j > \text{margin}$ selects the largest g_j . $\square$

Theorem 10.. Adding row v adds contrib_v to G and leaves u 's local terms unchanged, so $\Delta = \tilde{w}_u \text{contrib}_v = (a \cdot x)x - a \odot x^2 + a \odot c_v \odot x$. A logged DR entry satisfies $w \in [\hat{Y} - \hat{Y}/\tau, \hat{Y} + (1 - \hat{Y})/\tau]$ for any label and any $p \geq \tau$, and an unlogged one equals $\hat{Y}$. D is maximised (minimised) by assigning the L slots to the largest positive (negative) deviations. For fixed x_k the effect is linear in D , so its extremes are at the ends of the D interval; for fixed D it is quadratic in x_k , with extremes at the ends or at the clamped vertex. Effects add over fake users. For unclipped IPS, $|W_e| = 1/p_e$ is unbounded. $\square$

Theorem 11.. Because the population-level nuisances are public and every user-level part depends only on the user's own row, adding or removing user

v changes G by contrib_v alone. With the row scaling, $\|\text{contrib}_v\|_F \leq \|\tilde{w}_v \tilde{w}_v^\top - \text{diag}(\tilde{w}_v^2)\|_F + \|c_v^2 w_v\|_2 \leq R^2 + R^2$, so the ℓ_2 -sensitivity of the upper triangle is at most $2R^2$. The analytic Gaussian mechanism is (ϵ, δ) -DP for every $\epsilon > 0$ exactly when $\Phi(\frac{\Delta}{2\sigma} - \frac{\epsilon\sigma}{\Delta}) - e^\epsilon \Phi(-\frac{\Delta}{2\sigma} - \frac{\epsilon\sigma}{\Delta}) \leq \delta$ (Theorem 8 of Balle and Wang (2018)); we use the smallest such σ . The classical calibration $\sigma = \Delta \sqrt{2 \ln(1.25/\delta)}/\epsilon$ (Dwork and Roth, 2014) is only valid for $\epsilon < 1$. Each user’s output is a function of the release and the user’s own data, which gives joint DP by the billboard lemma (Hsu et al., 2014). $\square$

Theorem 12.. The allocation polytope is defined by the incidence matrix of the bipartite user–candidate graph with degree bounds, which is totally unimodular, so the linear relaxation has an integral optimal vertex and min-cost flow finds it. Rounding changes each score by at most $h/2$ with $h = (\max s - \min s)/L$. If x^* is optimal for s and $\hat{x}$ for the rounded scores $\tilde{s}$, then $\langle x^*, s \rangle \leq \langle x^*, \tilde{s} \rangle + NK h/2 \leq \langle \hat{x}, \tilde{s} \rangle + NK h/2 \leq \langle \hat{x}, s \rangle + NK h$, so $\Gamma \leq NK h$. Each item is recommended at most $\max_i \text{cap}_i$ times and NK slots are filled, which gives the coverage bound. With x_F optimal for U , $U(\hat{x}) \geq \langle \hat{x}, s \rangle - |\langle \hat{x}, s - F^* \rangle| \geq \langle x_F, s \rangle - \Gamma - |\langle \hat{x}, s - F^* \rangle| \geq U(x_F) - \Gamma - 2 \max_x |\langle x, s - F^* \rangle|$. If $\mathbb{E}s = F^*$, Theorem 6(d) applied to every candidate pair and a union bound give $|s_{ui} - F_{ui}^*| \leq t_\delta$ for all candidate pairs with probability $1 - \delta$, and an allocation sums NK such terms. $\square$

Appendix B. Experimental details

Training-free operators. All operators share the score $s = s_1/c_1 + \beta s_3/c_3$ (Section 4) with $\beta \in \{0, 0.03, 0.1, 0.3, 1, 3, 10\}$ or the three-hop term alone; the five-hop variants add $\gamma s_5/c_5$ with $\gamma \in \{0.03, 0.1, 0.3, 1, 3\}$. The grids are:

normalisation exponent $\alpha \in \{0.3, 0.5, 0.7\}$ (Coat), $\{0.5, 0.7\}$ (Yahoo! R3) and $\{0.3, 0.5\}$ (KuaiRec); clip $\tau \in \{0.01, 0.02, 0.05, 0.1, 0.2\}$ (Coat), $\{0.1, 0.2\}$ (Yahoo! R3) and $\{0.05, 0.1, 0.2\}$ (KuaiRec); imputation ridge $\lambda \in \{1, 5, 20\}$ (Coat), $\{20\}$ (Yahoo! R3) and $\{5, 20\}$ (KuaiRec); additive or low-rank (rank 32) imputation; control-variate weight $\in \{0, 0.25, 0.5, 1\}$ on Yahoo! R3 and 1 elsewhere; and degree source $\in \{\hat{Y}, \text{cross-fitted } W\}$ for the debiased operators. The degree source is a searched hyper-parameter; both options satisfy Remark 1, and only $\hat{Y}$ -degrees make the certificates of Theorem 10 informative. EASE uses $\lambda \in \{10, 100, 1000\}$; GF-CF uses the ideal low-pass weight $\in \{0, 0.3, 1\}$ with 64 or 256 singular vectors. On the DR graph, EASE and GF-CF search the same nuisance grid as DR adjacency. The propensity model is the one shipped with Coat and a logistic exposure model on user and item exposure rates elsewhere; on Yahoo! R3 it was selected on validation data against an outcome-dependent Naive-Bayes model. All nuisances are cross-fitted over ten random folds of pairs.

Trained models. Embeddings have 64 dimensions (Coat: 32 or 64) and LightGCN-type models two layers. The grid combines the learning rates 10^{-2} , $3 \cdot 10^{-3}$ and 10^{-3} with the weight decays 10^{-5} , 10^{-4} and 10^{-3} on Coat and Yahoo! R3, and the learning rates 10^{-2} and $3 \cdot 10^{-3}$ with the weight decays 10^{-5} and 10^{-4} on KuaiRec, plus $\gamma \in \{0.1, 0.2\}$ for PDA and $r \in \{0.3, 0.7\}$ for r-AdjNorm. BPR models draw one uniformly sampled negative per positive. Batch sizes are 256 (Coat), 1,024 (Yahoo! R3) and 32,768 (KuaiRec, with 2M sampled pairs per epoch for the pointwise losses); training runs for at most 100, 60 and 30 epochs with patience 5, 5 and 3 on the mean validation score. Because the training log is the same for every split, each configuration is trained once

and evaluated after every epoch on the validation part of every split; each split then selects its own configuration and epoch. The split-0 selection is re-trained with three (KuaiRec: two) further seeds to measure seed variance. Table B.13 lists the number of configurations each method was selected from.

Evaluation and tests. Coat splits each user’s candidates (30% validation), Yahoo! R3 and KuaiRec split the test users (30% validation), over 10, 5 and 5 random splits. Users without a positive candidate are skipped. For a pair of methods, each test user’s metric difference is averaged over the splits in which the user is in the test part, and we report the mean difference over users with a 95% t -interval and the paired t -test, Holm-adjusted over all comparisons of a dataset. The Wilcoxon signed-rank test gives the same conclusions except for the five-hop DRUP against DR comparison on KuaiRec ($p = 0.66$ against 0.011). The corrected resampled t -test over split means (Nadeau and Bengio, 2003) is far more conservative with five or ten splits and is reported in the result files.

Hardware and code. All experiments ran on four CPU cores with 16 GB of memory, without a GPU. Code, configurations and result files are available at the repository given on the title page.

Table 3: Accuracy on unbiased test data (mean $\pm$ std over splits). Best in bold, second underlined. “–”: not run (five hops only where the three-hop DR operators are competitive). Every method is tuned per split on the same validation data; Table B.13 lists the number of configurations.

Method	Coat nDCG@5	Yahoo! R3 nDCG@5	KuaiRec nDCG@20
<i>Non-personalised</i>			
Popularity	0.5367 $\pm$ 0.0095	0.5331 $\pm$ 0.0059	0.5766 $\pm$ 0.0023
Imputation only	0.5532 $\pm$ 0.0097	0.5162 $\pm$ 0.0044	0.6194 $\pm$ 0.0018
<i>Trained, pointwise loss</i>			
MF	0.4766 $\pm$ 0.0166	0.5076 $\pm$ 0.0060	0.6227 $\pm$ 0.0029
IPS-MF	0.4691 $\pm$ 0.0163	0.5098 $\pm$ 0.0048	0.6258 $\pm$ 0.0025
DR-MF	0.5259 $\pm$ 0.0132	0.5116 $\pm$ 0.0020	0.6191 $\pm$ 0.0020
LightGCN	0.5261 $\pm$ 0.0123	0.5935 $\pm$ 0.0059	0.6277 $\pm$ 0.0027
DR-LightGCN	0.5500 $\pm$ 0.0168	0.5440 $\pm$ 0.0029	0.6246 $\pm$ 0.0016
<i>Trained, pairwise loss (BPR)</i>			
MF	0.5133 $\pm$ 0.0211	0.6460 $\pm$ 0.0041	0.5913 $\pm$ 0.0028
PDA	0.5198 $\pm$ 0.0160	0.6516 $\pm$ 0.0052	0.5822 $\pm$ 0.0020
LightGCN	<u>0.5703$\pm$0.0141</u>	0.6693 $\pm$ 0.0045	0.6086 $\pm$ 0.0028
r-AdjNorm	0.5746$\pm$0.0132	0.6713$\pm$0.0050	0.6086 $\pm$ 0.0031
NAVIP	0.5665 $\pm$ 0.0133	<u>0.6694$\pm$0.0059</u>	0.6183 $\pm$ 0.0024
<i>Training-free, logged graph</i>			
linear LightGCN / r-AdjNorm	0.5571 $\pm$ 0.0100	0.6585 $\pm$ 0.0058	0.6036 $\pm$ 0.0028
EASE	0.5401 $\pm$ 0.0138	0.6478 $\pm$ 0.0061	0.5213 $\pm$ 0.0032
GF-CF	0.5512 $\pm$ 0.0124	0.6569 $\pm$ 0.0051	0.6141 $\pm$ 0.0025
<i>Training-free, debiased graph</i>			
IPS adjacency (NAVIP-style)	0.5510 $\pm$ 0.0224	0.6567 $\pm$ 0.0063	0.6132 $\pm$ 0.0030
IPS + walk correction	0.5510 $\pm$ 0.0224	0.6567 $\pm$ 0.0063	0.6132 $\pm$ 0.0030
EASE on DR graph	0.5564 $\pm$ 0.0141	0.6072 $\pm$ 0.0033	0.6026 $\pm$ 0.0022
GF-CF on DR graph	0.5598 $\pm$ 0.0084	0.6313 $\pm$ 0.0051	0.6371 $\pm$ 0.0021
DR adjacency	0.5459 $\pm$ 0.0244	0.6567 $\pm$ 0.0063	<u>0.6378$\pm$0.0020</u>
DRUP	0.5467 $\pm$ 0.0235	0.6567 $\pm$ 0.0063	0.6378 $\pm$ 0.0019
DR adjacency, 5 hops	0.5524 $\pm$ 0.0245	–	0.6379$\pm$0.0011
DRUP , 5 hops	0.5451 $\pm$ 0.0204	–	0.6376 $\pm$ 0.0011

Table 4: User-level paired comparisons of DRUP with each method: mean difference of the test metric over users with a 95% interval, and the t -test p -value, Holm-adjusted over all comparisons of the dataset (225, 2,310 and 1,410 users); selected rows, all comparisons are in the result files. The first two rows isolate the walk correction.

DRUP against	Coat		Yahoo! R3		KuaiRec	
	Δ [95% CI]	p_{Holm}	Δ [95% CI]	p_{Holm}	Δ [95% CI]	p_{Holm}
DR adjacency (effect of the correction)	+0.0014 [-0.002, +0.005]	1.000	+0.0000 [+0.000, +0.000]	1.000	-0.0001 [-0.000, +0.000]	0.454
DR adjacency (effect of the correction), both 5 hops	—	—	—	—	-0.0003 [-0.001, -0.000]	0.043
linear LightGCN on the logged graph	-0.0123 [-0.051, +0.027]	1.000	-0.0022 [-0.005, +0.001]	1.000	+0.0346 [+0.030, +0.039]	$< 10^{-3}$
imputation only	-0.0082 [-0.017, +0.001]	1.000	+0.1411 [+0.125, +0.157]	$< 10^{-3}$	+0.0179 [+0.016, +0.020]	$< 10^{-3}$
GF-CF	-0.0069 [-0.046, +0.032]	1.000	-0.0018 [-0.008, +0.005]	1.000	+0.0243 [+0.021, +0.028]	$< 10^{-3}$
GF-CF on DR graph	-0.0129 [-0.021, -0.005]	0.032	+0.0248 [+0.018, +0.032]	$< 10^{-3}$	+0.0006 [-0.001, +0.002]	1.000
LightGCN (pointwise)	+0.0202 [-0.002, +0.042]	1.000	+0.0628 [+0.049, +0.076]	$< 10^{-3}$	+0.0098 [+0.007, +0.013]	$< 10^{-3}$
LightGCN (BPR)	-0.0274 [-0.061, +0.006]	1.000	-0.0125 [-0.021, -0.004]	0.039	+0.0295 [+0.026, +0.033]	$< 10^{-3}$
r-AdjNorm	-0.0320 [-0.066, +0.002]	1.000	-0.0149 [-0.023, -0.007]	0.004	+0.0292 [+0.026, +0.033]	$< 10^{-3}$
NAVIP	-0.0238 [-0.057, +0.010]	1.000	-0.0127 [-0.021, -0.004]	0.031	+0.0197 [+0.017, +0.023]	$< 10^{-3}$
MF (BPR)	+0.0346 [+0.004, +0.065]	0.499	+0.0121 [+0.003, +0.021]	0.077	+0.0463 [+0.042, +0.051]	$< 10^{-3}$
PDA	+0.0251 [-0.008, +0.058]	1.000	+0.0054 [-0.004, +0.015]	1.000	+0.0549 [+0.050, +0.060]	$< 10^{-3}$

Table 5: KuaiRec with the log thinned to a fraction q of its exposures (nuisances re-fitted, additive imputation, $\tau \in \{0.05, 0.01\}$, $\lambda \in \{0, 1\}$, selected on validation). nDCG@20 of the uncorrected and corrected operators, and the lowest top-20 overlap and Spearman correlation between the two rankings at matched configurations.

thinning q	density	DR	DRUP	DR 5 hops	DRUP 5 hops	top-20 overlap	Spearman
1	13.38%	0.6302	0.6302	0.6306	0.6306	0.971	0.999
0.1	1.34%	0.6097	0.6102	0.6109	0.6111	0.935	0.999
0.03	0.40%	0.6073	0.6069	0.6085	0.6081	0.985	1.000

Table 6: Real-data intervention $do(p \leftarrow p/2)$: the logged exposures of a random half of the items are thinned with probability 1/2 (three to five draws). Rank shift: change of the treated items’ mean within-user rank relative to control items (0: invariant). η : elasticity of the mean score of treated relative to control candidates (1: exposure-proportional, 0: invariant). Nuisances are frozen, re-fitted with the known propensity change, or re-fitted with propensities re-estimated on the thinned log. One configuration per operator (the one selected most often), all test data.

Data	Operator	frozen nuisances		re-fitted, known $p/2$		re-fitted, re-estimated $\hat{p}$	
		rank shift	η	rank shift	η	rank shift	η
Coat	logged graph	-0.0670	+1.031	-0.0585	+0.748	–	–
	IPS adjacency	-0.0652	+0.899	-0.0651	+0.900	–	–
	DR adjacency	+0.0021	-0.005	+0.0001	-0.019	–	–
	DRUP	+0.0036	-0.008	+0.0015	-0.022	–	–
	treated logged pairs below the DRUP clip after $do(p \leftarrow p/2)$: 94%						
Yahoo! R3	logged graph	-0.1611	+1.002	-0.0943	+0.508	-0.0943	+0.508
	IPS adjacency	-0.1504	+0.959	-0.0864	+0.481	-0.0873	+0.485
	DR adjacency	-0.1504	+0.959	-0.0864	+0.481	-0.0873	+0.485
	DRUP	-0.1504	+0.959	-0.0864	+0.481	-0.0873	+0.485
	treated logged pairs below the DRUP clip after $do(p \leftarrow p/2)$: 95%						
KuaiRec	logged graph	-0.2408	+0.997	-0.1339	+0.500	-0.1339	+0.500
	IPS adjacency	-0.0462	+0.137	-0.0466	+0.141	-0.0657	+0.218
	DR adjacency	+0.0061	-0.013	+0.0001	-0.003	+0.0017	-0.006
	DRUP	+0.0057	-0.012	-0.0004	-0.002	+0.0011	-0.005
	treated logged pairs below the DRUP clip after $do(p \leftarrow p/2)$: 16%						

Table 7: Clip τ against exposure invariance for DRUP under the same intervention, with frozen nuisances (η : mean $\pm$ std over draws).

Data	Operator, τ	nDCG	rank shift	η	logged pairs clipped
Coat	logged graph	0.4917	-0.0646	+1.034 $\pm$ 0.043	–
	DRUP, $\tau = 0.01$	0.4191	+0.0071	-0.021 $\pm$ 0.008	1%
	DRUP, $\tau = 0.02$	0.4357	+0.0187	-0.054 $\pm$ 0.013	6%
	DRUP, $\tau = 0.05$	0.4599	+0.0104	-0.033 $\pm$ 0.008	12%
	DRUP, $\tau = 0.1$	0.4805	+0.0068	-0.021 $\pm$ 0.004	25%
	DRUP, $\tau = 0.2$	0.4891	+0.0046	-0.009 $\pm$ 0.002	77%
Yahoo! R3	logged graph	0.6590	-0.1604	+1.000 $\pm$ 0.011	–
	DRUP, $\tau = 0.0005$	0.6136	-0.0477	+0.181 $\pm$ 0.113	0%
	DRUP, $\tau = 0.002$	0.6140	-0.0519	+0.195 $\pm$ 0.099	1%
	DRUP, $\tau = 0.01$	0.6239	-0.0878	+0.508 $\pm$ 0.055	15%
	DRUP, $\tau = 0.05$	0.6500	-0.1274	+0.825 $\pm$ 0.017	51%
	DRUP, $\tau = 0.1$	0.6570	-0.1394	+0.901 $\pm$ 0.013	67%
	DRUP, $\tau = 0.2$	0.6572	-0.1494	+0.959 $\pm$ 0.009	82%
KuaiRec	logged graph	0.6026	-0.2412	+0.997 $\pm$ 0.003	–
	DRUP, $\tau = 0.01$	0.5309	-0.0000	-0.002 $\pm$ 0.002	2%
	DRUP, $\tau = 0.05$	0.6353	+0.0033	-0.007 $\pm$ 0.001	7%
	DRUP, $\tau = 0.1$	0.6382	+0.0059	-0.012 $\pm$ 0.000	10%
	DRUP, $\tau = 0.2$	0.6359	+0.0099	-0.021 $\pm$ 0.000	16%

Table 8: Exposure concentration and user groups. PRU: popularity–rank correlation. ECB (KuaiRec, where item quality is known from the fully observed matrix): partial correlation between an item’s mean rank and its log exposure, controlling for its true like rate. User gap: nDCG of group 0 minus group 1 (men minus women on Coat, inactive minus active users elsewhere) with a bootstrap 95% interval.

Data	Operator	nDCG	PRU	ECB	Gini@ K	user gap [95% CI]	coverage
Coat	logged graph	0.4917	0.394	–	0.340	+0.112 [+0.033, +0.190]	0.937
	IPS adjacency	0.4951	0.415	–	0.346	+0.108 [+0.027, +0.192]	0.937
	DR adjacency	0.4890	0.689	–	0.599	+0.111 [+0.024, +0.189]	0.667
	DRUP	0.4891	0.688	–	0.600	+0.111 [+0.025, +0.189]	0.663
Yahoo!R3	logged graph	0.6590	0.534	–	0.281	-0.046 [-0.071, -0.019]	1.000
	IPS adjacency	0.6572	0.515	–	0.272	-0.044 [-0.069, -0.018]	1.000
	DR adjacency	0.6572	0.515	–	0.272	-0.044 [-0.069, -0.018]	1.000
	DRUP	0.6572	0.515	–	0.272	-0.044 [-0.069, -0.018]	1.000
KuaiRec	logged graph	0.6026	0.845	0.830	0.991	+0.008 [-0.012, +0.029]	0.152
	IPS adjacency	0.6123	0.875	0.801	0.992	+0.014 [-0.007, +0.035]	0.081
	DR adjacency	0.6385	0.802	0.329	0.994	+0.012 [-0.009, +0.031]	0.012
	DRUP	0.6382	0.803	0.331	0.994	+0.012 [-0.009, +0.031]	0.012

Table 9: Exposure-capped re-ranking (Theorem 12, exact min-cost flow). Each item may be recommended to at most c times its fair share NK/n of users. Cells: $\text{nDCG}@K$ ($\text{Gini}@K$). Every cap held; the full range of caps and the accuracy–concentration frontier are in the Supplementary Material.

Data	Operator	no cap	$c = 5$	$c = 2$
Coat	logged graph	0.502 (0.34)	0.501 (0.34)	0.494 (0.32)
	IPS adjacency	0.507 (0.35)	0.503 (0.34)	0.493 (0.33)
	DR adjacency	0.489 (0.60)	0.489 (0.60)	0.475 (0.48)
	DRUP	0.489 (0.60)	0.489 (0.60)	0.476 (0.48)
Yahoo! R3	logged graph	0.658 (0.28)	0.658 (0.28)	0.658 (0.27)
	IPS adjacency	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)
	DR adjacency	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)
	DRUP	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)
KuaiRec	logged graph	0.603 (0.99)	0.173 (0.72)	0.115 (0.44)
	IPS adjacency	0.612 (0.99)	0.178 (0.79)	0.121 (0.49)
	DR adjacency	0.639 (0.99)	0.199 (0.80)	0.140 (0.50)
	DRUP	0.638 (0.99)	0.199 (0.80)	0.140 (0.50)

Table 10: Exact explanations (Proposition 8, Theorem 9) on 200 random test users: largest relative error of completeness and subset additivity against re-propagation; rank of the top-1 item after un-logging the five interactions with the largest attribution, five random ones, or the five most similar ones (0 = still first); share of top-1 recommendations for which no subset of the user’s own interactions makes the second item overtake it; median size of the minimal counterfactual explanation otherwise. Every explanation was verified by re-scoring.

Data	max. rel. error	top-1 rank after 5 removals (attr. / random / similarity)	no CF explanation	median CF size
Coat	6.1e-16	0.04 / 0.01 / 0.01	94%	2.0
Yahoo! R3	8.7e-16	1.65 / 0.38 / 1.40	10%	1.0
KuaiRec	3.1e-06	0.03 / 0.01 / 0.01	52%	7.0

Table 11: Injection attack: F fake users log a low-popularity target and the most popular items. Realised hit rate of the target in real users’ top- K , and the fraction of real users for whom Theorem 10 certifies that the target cannot enter the top- K against *any* F profiles of that size (frozen nuisances). No certificate was violated.

Data	Operator	realised hit rate			certified fraction		
Coat	$F =$	1	5	20	1	5	20
	logged graph	0.004	0.009	0.444	0.00	0.00	0.00
	IPS adjacency	0.001	0.269	0.661	0.15	0.00	0.00
	DR adjacency	0.000	0.000	0.029	1.00	0.94	0.11
	DRUP	0.000	0.000	0.028	1.00	0.93	0.10
Yahoo! R3	$F =$	1	5	100	1	5	100
	logged graph	0.065	0.065	0.199	0.12	0.00	0.00
	IPS adjacency	0.065	0.065	0.171	0.00	0.00	0.00
	DR adjacency	0.065	0.065	0.171	0.00	0.00	0.00
	DRUP	0.065	0.065	0.171	0.00	0.00	0.00
KuaiRec	$F =$	1	5	200	1	5	200
	logged graph	0.002	0.002	0.288	0.00	0.00	0.00
	IPS adjacency	0.001	0.001	0.105	0.98	0.61	0.00
	DR adjacency	0.000	0.000	0.393	1.00	0.95	0.00
	DRUP	0.000	0.000	0.393	1.00	0.95	0.00

Table 12: nDCG of jointly private DRUP ($\delta = 10^{-5}$, analytic Gaussian mechanism, denoising rank 32 fixed in advance). Population-level nuisances come from a public 10% of the users (non-test users where available; on Coat, whose users are all test users, the public users are excluded from evaluation); “all nuisances” is the non-private model with all nuisances fitted on the whole log. Other ranks in Table S6.

Data	$\epsilon = 0.5$	1	2	4	8	16	∞	all nuisances	popularity
Coat	0.391	0.402	0.416	0.434	0.454	0.457	0.454	0.555	0.527
Yahoo! R3	0.397	0.437	0.480	0.518	0.551	0.574	0.608	0.652	0.525
KuaiRec	0.049	0.052	0.055	0.163	0.530	0.596	0.635	0.637	0.577

Table B.13: Number of configurations searched per method (per split).

Method	Coat	Yahoo! R3	KuaiRec
Popularity (non-personalised)	1	1	1
Imputation only (non-personalised)	6	2	4
MF (trained)	18	9	4
IPS-MF (trained)	18	9	4
DR-MF (trained)	18	9	4
LightGCN (trained)	18	9	4
DR-LightGCN (trained)	18	9	4
MF (trained)	18	9	4
PDA (trained)	36	18	8
LightGCN (trained)	18	9	4
r-AdjNorm (trained)	36	18	8
NAVIP (trained)	18	9	4
linear LightGCN / r-AdjNorm (training-free)	24	16	16
EASE (training-free)	3	3	3
GF-CF (training-free)	4	4	4
IPS adjacency (NAVIP-style) (training-free)	720	64	192
IPS + walk correction (training-free)	720	64	192
EASE on DR graph (training-free)	90	48	36
GF-CF on DR graph (training-free)	120	64	48
DR adjacency (training-free)	1440	512	384
DRUP (training-free)	1440	512	384
DR adjacency, 5 hops (training-free)	6300	–	420
DRUP , 5 hops (training-free)	6300	–	420